

JavaScript Control Flow

Control Flow determines **how your code executes** — the order in which statements run in a program.

By default, JavaScript executes code **from top to bottom, left to right**.

However, using **control flow statements**, we can make decisions and repeat actions based on conditions.

1. Conditional Statements

Conditional statements let the program **make decisions** based on certain conditions (true/false values).

1.1 if Statement

Used to run a block of code **only if** a condition is true.

```
let age = 18;

if (age >= 18) {
  console.log("You are an adult.");
}
```

✓ Explanation:

If the condition inside parentheses `()` is **true**, the code inside `{}` runs.

If it's **false**, JavaScript skips it.

1.2 if...else Statement

Used when there are **two possible outcomes**.

```
let age = 16;

if (age >= 18) {
  console.log("You can vote.");
} else {
  console.log("You are too young to vote.");
}
```

✓ Explanation:

- If condition is true → if block runs
- Else → else block runs

1.3 if...else if...else Statement

Used for **multiple conditions**.

```
let score = 75;

if (score >= 90) {
  console.log("Excellent");
} else if (score >= 70) {
  console.log("Good");
} else if (score >= 50) {
  console.log("Fair");
} else {
  console.log("Fail");
}
```

✓ Explanation:

- Checks conditions **in order**.
- As soon as one is true, the others are ignored.

1.4 switch Statement

Used when checking **one variable against multiple possible values**.

```
let day = 3;

switch (day) {
  case 1:
    console.log("Monday");
    break;
  case 2:
    console.log("Tuesday");
    break;
  case 3:
    console.log("Wednesday");
    break;
  default:
    console.log("Invalid day");
}
```

✓ Notes:

- Each `case` represents a possible value.
- Use `break` to stop the switch after a match.
- `default` runs if no case matches.

2. Loops (Repetition Statements)

Loops allow you to **repeat a block of code** multiple times, either for a specific count or while a condition is true.

2.1 for Loop

Runs code a specific number of times.

```
for (let i = 0; i < 5; i++) {
  console.log("Count:", i);
}
```

✓ Explanation:

- `let i = 0` → start
- `i < 5` → condition

- `i++` → increment after each loop
- Stops when condition becomes false.

2.2 while Loop

Runs code **while** a condition is true.

```
let i = 0;
while (i < 5) {
  console.log(i);
  i++;
}
```

✓ Notes:

- Checks the condition **before** running the loop.
- If condition is false at the start, it won't run at all.

2.3 do...while Loop

Similar to `while`, but **runs at least once**, even if condition is false.

```
let i = 0;
do {
  console.log(i);
  i++;
} while (i < 5);
```

✓ Notes:

- Condition is checked **after** executing the code block.

2.4 for...of Loop

Used to loop through **arrays or iterable objects**.

```
let fruits = ["apple", "banana", "cherry"];

for (let fruit of fruits) {
  console.log(fruit);
}
```

✓ Notes:

- Loops through **values** in an array.

2.5 for...in Loop

Used to loop through **object properties**.

```
let person = { name: "Ismail", age: 30 };

for (let key in person) {
  console.log(key, person[key]);
}
```

✓ Notes:

- Loops through **keys (property names)** of an object.

3. Jump Statements

Used to control the flow inside loops.

3.1 break Statement

Stops the loop entirely.

```
for (let i = 1; i <= 5; i++) {
  if (i === 3) break;
  console.log(i);
}

// Output: 1, 2
```

3.2 continue Statement

Skips the current iteration and moves to the next.

```
for (let i = 1; i <= 5; i++) {
  if (i === 3) continue;
  console.log(i);
}
// Output: 1, 2, 4, 5
```

4. Summary Table

Type	Statement	Purpose	Example
Condition	if / else / switch	Make decisions	Run different code based on conditions
Loop	for , while , do...while	Repeat code	Run code multiple times
Loop Control	break , continue	Manage loop execution	Stop or skip iteration

5. Example: Combined Control Flow

```
let students = ["John", "Mary", "Ismail", "Tunde"];

for (let student of students) {
  if (student === "Ismail") {
    console.log("Admin found!");
    break;
  }
  console.log("Checking:", student);
}
```